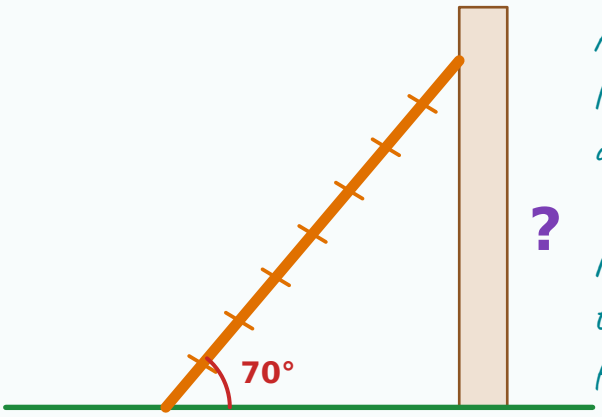
Chapter 28

Introduction to trigonometry

triangles that measure what you can't reach



A 5 m ladder leans at 70°. How high up the wall does it reach?

No tape measure needed: trigonometry answers it from the angle alone.

Trigonometry means "triangle measuring". Surveyors use it to find the height of mountains, sat-navs to find where you are, engineers to find forces. It all rests on one fact about right-angled triangles: **the angle decides the shape**.

We'll name the sides properly, meet Pythagoras (with real squares), see **why** sin, cos and tan exist, and work Lecture 6's examples one tiny step at a time.

Goes with **Lecture 6** (right-angled triangles, Pythagoras in 2D and 3D, the circle, SOHCAHTOA, inverse functions, Examples 1-3).

Naming the sides: stand at the angle and look

RULE: naming the sides of a right-angled triangle

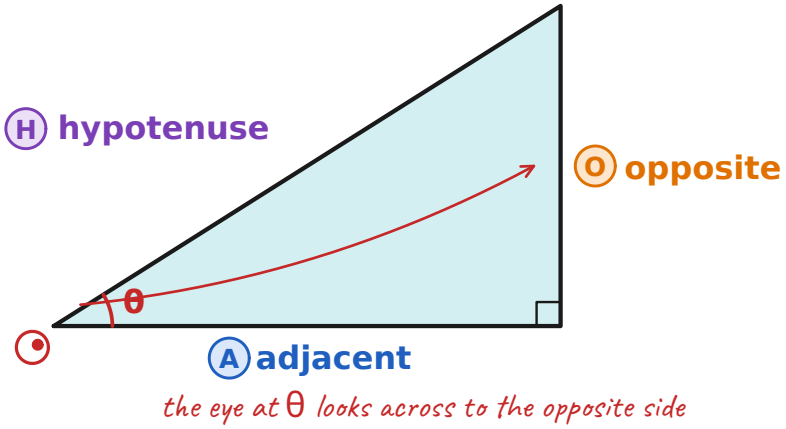
Hypotenuse (H): the longest side, opposite the right angle. It never changes.

Opposite (O): the side across from the angle θ you are using.

Adjacent (A): the side next to θ that is not the hypotenuse.

O and A depend on which angle is θ: change the angle and they swap.

A right-angled triangle has one 90° corner (the little square). Its three sides get names, and two of the names depend on **which angle you're standing at**, called θ ("theta", a Greek letter used for angles).

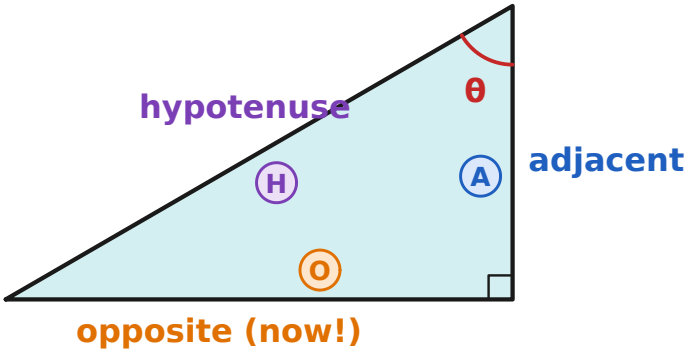


HYPOTENUSE: the longest side, opposite the right angle. It never changes name.

OPPOSITE: across from θ, the side θ is "looking at".

ADJACENT: next to θ ("adjacent" means next to), but not the hypotenuse.

Stand at the other corner and opposite and adjacent **swap**; the hypotenuse stays put:



Always label from θ's point of view, one side at a time:
1. hypotenuse (easy, longest)
2. opposite (across from θ)
3. adjacent (the one left over)

Lecture 6 writes them h, o and a.

The labelling habit. From here on, every triangle in this book's trig chapters wears three badges: **H** (hypotenuse, purple), **O** (opposite, orange) and **A** (adjacent, blue), always in the same colours. Do the same on paper: **mark θ, then badge all three sides, before anything else**, even when a question only needs two of them. After a week it becomes automatic, and choosing SOH, CAH or TOA becomes a matter of reading your own badges.

◿ Pythagoras: squares on the sides

RULE: Pythagoras' theorem

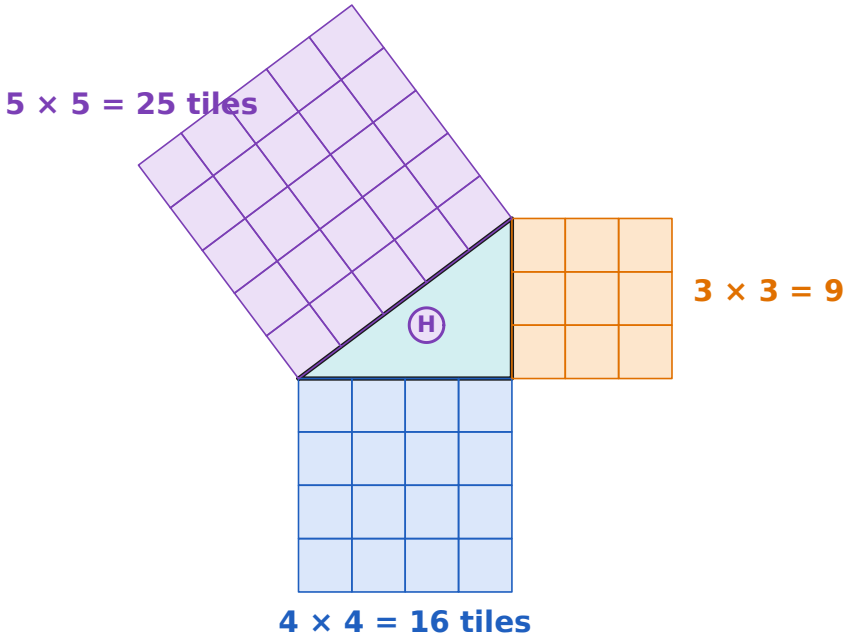
In a **right-angled** triangle with hypotenuse c and shorter sides a and b: $a^2 + b^2 = c^2$.

Finding the hypotenuse: square the two short sides, **add**, square-root: $c = \sqrt{a^2 + b^2}$.

Finding a short side: square both, **subtract** (big minus small), square-root: $a = \sqrt{c^2 - b^2}$.

Only for right-angled triangles. Other triangles need Chapter 31's cosine rule.

Build a real square on each side of a 3-4-5 triangle out of 1 × 1 tiles and count them:



9 + 16 = 25. The two small squares exactly fill the big one.

True for every right-angled triangle.

$$h^2 = o^2 + a^2$$ ← *Pythagoras: hypotenuse squared = the other two squared, added*

Finding a short side means subtracting instead. Lecture 6, Example 3: $h = 5$, $o = 2.5$:

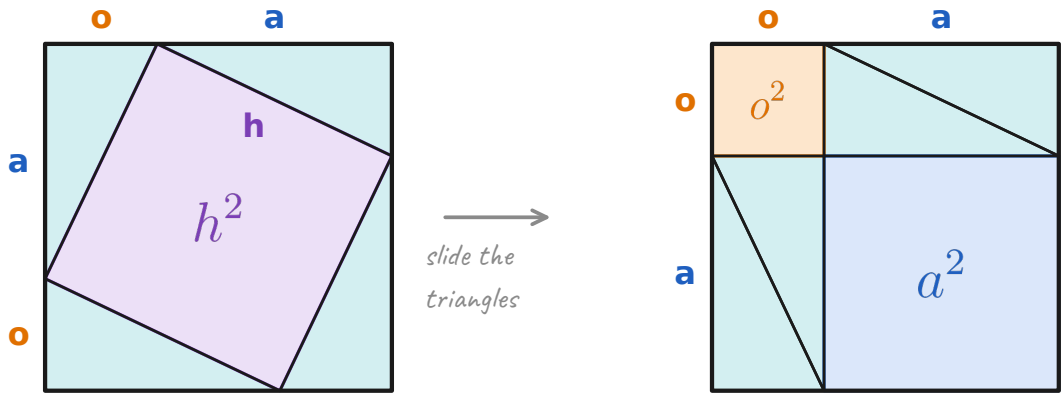
$$a^2 = 5^2 - 2.5^2 = 25 - 6.25 = 18.75$$

$$a = \sqrt{18.75} \approx 4.33$$ ← don't forget the square root!

(The slide writes "$5^2 - 2.5^2 = a = 4.33$", dropping the square root: $5^2 - 2.5^2$ is 18.75, not 4.33.)

Why it's always true: four triangles in a box

Counting tiles only checks one triangle. Here's why it works for **every** right-angled triangle. Take four copies of any one, with short sides o and a and hypotenuse h, and a square box with sides $o + a$. Pack the triangles in two different ways:



4 triangles, and the tilted square on the hypotenuse

the same 4 triangles, paired into 2 rectangles, and two squares

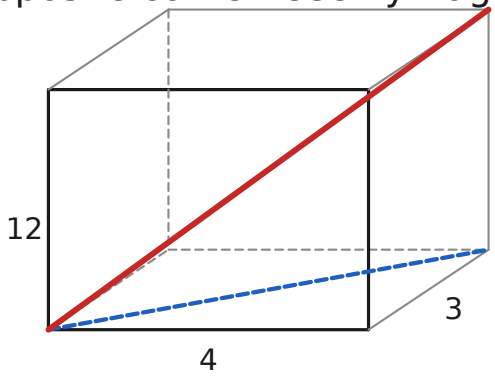
Both boxes are the same size and hold the same four triangles. So the space the triangles leave must be the same too: on the left that's h^2, on the right it's $o^2 + a^2$. Nothing here depended on the lengths, so it's true for **every** right-angled triangle:

$$h^2 = o^2 + a^2$$ ← the leftover space, counted two ways

(Chapter 11's perfect square is hiding here: the whole box is $(o + a)^2 = o^2 + 2oa + a^2$, and the four triangles are the two oa rectangles.)

◿ Pythagoras in 3D, and the circle

A box: the longest stick that fits in a 3 × 4 × 12 box goes corner to opposite corner. Use Pythagoras twice: across the floor, then up:



floor: $\sqrt{4^2 + 3^2} = 5$

stick: $\sqrt{5^2 + 12^2} = 13$

$$d^2 = a^2 + b^2 + c^2$$

A circle is every point at distance r from the centre (a, b). The distance from the centre to a point (x, y) is the hypotenuse of a triangle with sides $x - a$ and $y - b$, so Pythagoras gives Lecture 6's circle equation:

$$(x - a)^2 + (y - b)^2 = r^2$$ ← centred at the origin: $x^2 + y^2 = r^2$

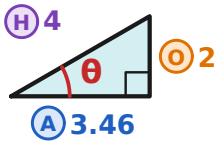
Why sin, cos and tan exist: zoomed photos

RULE: SOHCAHTOA

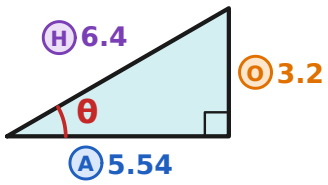
$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \text{ (SOH) } \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \text{ (CAH) } \tan\theta = \frac{\text{opposite}}{\text{adjacent}} \text{ (TOA)}$$

Each ratio depends **only on the angle**, not on the size of the triangle. Right-angled triangles only.

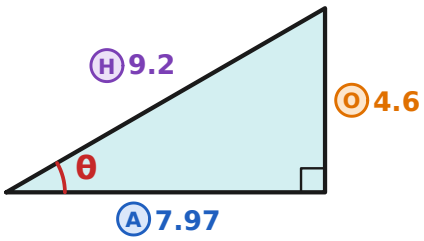
Draw a right-angled triangle with a 30° angle. Now photograph it and zoom in: the triangle gets bigger but its **shape** can't change, because the angles are the same. Measure the sides of each copy:



O ÷ H: 2 ÷ 4 = 0.5



O ÷ H: 3.2 ÷ 6.4 = 0.5



O ÷ H: 4.6 ÷ 9.2 = 0.5

Every 30° triangle has opposite ÷ hypotenuse = 0.5, however big. The ratio depends **only on the angle**. So we give each ratio a name and let the calculator remember them:

S O H

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

(sine)

C A H

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$

(cosine)

T O A

$$\tan\theta = \frac{\text{opp}}{\text{adj}}$$

(tangent)

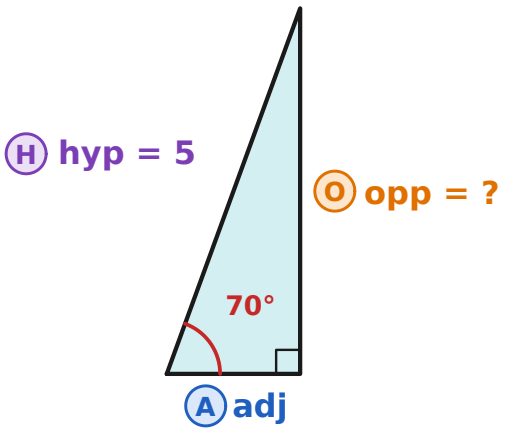
SOHCAHTOA glues the codes together. Lecture 6 admits it's a nightmare to pronounce and suggests your own sentence, e.g. *"Some Old Horses Can Always Hear Their Owners Approach"* . Degrees or radians doesn't matter here: like measuring in feet or inches.

◿ The ladder, in five tiny steps

PROCEDURE: which ratio? (sin, cos or tan)

1. Mark the angle θ you know (or want) and the right angle.
2. Label the sides H, O, A from θ's point of view.
3. Tick the side you **know** and the side you **want**.
4. Pick the ratio that contains exactly those two: O and H → **sin**; A and H → **cos**; O and A → **tan**.
5. Write the equation and solve it. Unknown on top: **multiply** ($\mathrm{O} = \mathrm{H}\sin\theta$). Unknown on the bottom: **divide** ($\mathrm{H} = \mathrm{O} \div \sin\theta$). Unknown angle: use the **inverse** ($\theta = \sin^{-1}\frac{\mathrm{O}}{\mathrm{H}}$).

A 5 m ladder makes 70° with the ground. How high up the wall does it reach?



1 Label the sides from θ = 70°: the ladder is the hypotenuse, the wall height is opposite.

2 What do I know, what do I want? Know hyp, want opp: O and H, so SOH, sine.

3 Write it: sin 70° = opp ÷ 5.

4 Transpose (Chapter 14): opp = 5 × sin 70°.

5 Calculate: 5 × 0.9397 = 4.70 m.

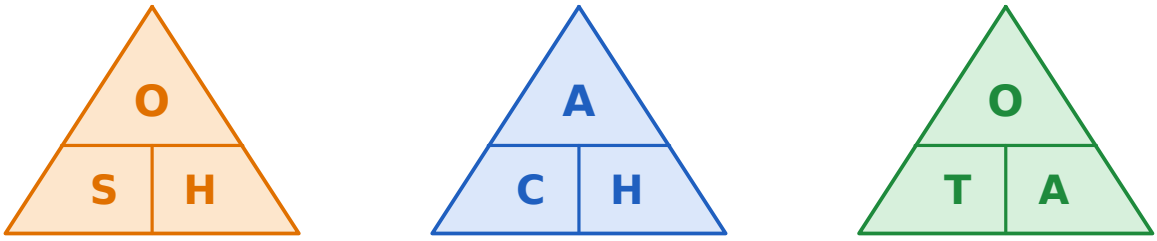
How far is its foot from the wall? Now we want the adjacent side: A and H, so CAH, cosine:

$$\cos 70^\circ = \frac{\mathrm{adj}}{5}$$

$$\mathrm{adj} = 5\cos 70^\circ = 5 \times 0.342 = 1.71 \text{ m}$$

Check with Pythagoras: $4.70^2 + 1.71^2 = 22.09 + 2.92 = 25.0 = 5^2$ ✓.

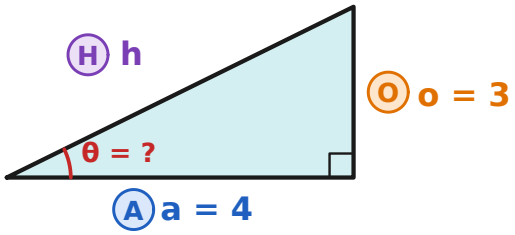
Cover-up triangles save the transposing. Put the top letter over the other two; cover what you want:



Cover O in the first: O = S × H. Cover H: H = O ÷ S. (Same idea as the speed-distance-time triangle.)

Lecture 6's examples, and the undo button

Example 1: opposite = 3, adjacent = 4. Find θ.

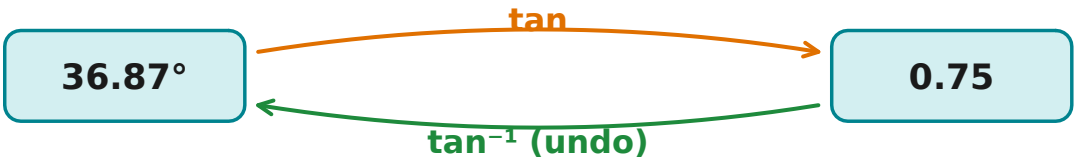


badges: O and A are known, so TOA (tan)

$$\tan\theta = \frac{3}{4} = 0.75$$

Now θ is **stuck inside** the tan. We need an **undo button** for tan, just as in Chapter 14 (÷ undoes ×) and Chapter 22 (log undoes power). It's called **inverse tan**, written $\tan^{-1}$ (or arctan):

$$\theta = \tan^{-1}(0.75) = 36.87^\circ$$ ← *SHIFT then tan on the calculator*



Warning: the -1 here means "undo", **not** "1 over" (Chapter 9's negative powers). $\sin^{-1} x$ is not $\frac{1}{\sin x}$. Sorry, mathematicians reused the symbol.

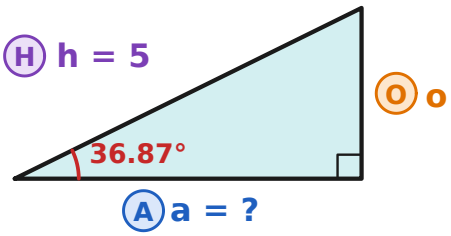
RULE: inverse trig gives ONE angle (the principal value)

$\sin^{-1}$ always answers between $-90°$ and $90°$. $\cos^{-1}$ between $0°$ and $180°$. $\tan^{-1}$ between $-90°$ and $90°$.

In a right-angled triangle that is the angle you want: it is always between $0°$ and $90°$.

But other angles share the same ratio: $\sin 150° = \sin 30° = 0.5$, yet $\sin^{-1}(0.5)$ only says $30°$. The other answers come from the quadrant rules (Chapter 29) and equation solving (Chapter 30).

Example 2: $h = 5$, $\theta = 36.87°$. Find a.

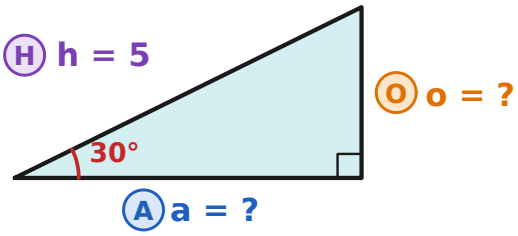


badges: know H, want A, so CAH (cos)

$$a = 5 \times \cos 36.87° = 4$$

(Lecture 6 asks "what do we think about the notation here?": it writes 36.870° with a superscript o. Use a proper ° sign.)

Example 3: $h = 5$, $\theta = 30°$. Find all the sides.



badges: know H, want O first, so SOH (sin); then Pythagoras for A

$$o = 5 \sin 30° = 2.5, \qquad a = \sqrt{5^2 - 2.5^2} = 4.33$$

Elevation, depression, and two linked triangles

RULE: angles of elevation and depression

Angle of elevation: measured **up** from the horizontal to the line of sight (looking up).

Angle of depression: measured **down** from the horizontal to the line of sight (looking down).

The angle of depression from A down to B equals the angle of elevation from B up to A (the two horizontal lines are parallel).

Both are measured from the **horizontal**, never from the vertical.

PROCEDURE: multi-step problems (two linked triangles)

1. Draw the picture and split it into right-angled triangles. Mark every known length and angle.
2. Find the side the two triangles **share**, using the triangle you know enough about.
3. Use that shared side in the second triangle to get the answer.
4. Keep full calculator accuracy in the middle; round only at the end.

Example 1 (elevation): from 40 m away, the angle of elevation of the top of a building is 35°. How tall is the building?

$$\tan 35^\circ = \frac{h}{40} \Rightarrow h = 40 \tan 35^\circ = 28.01 \text{ m}$$

know A, want O: TOA

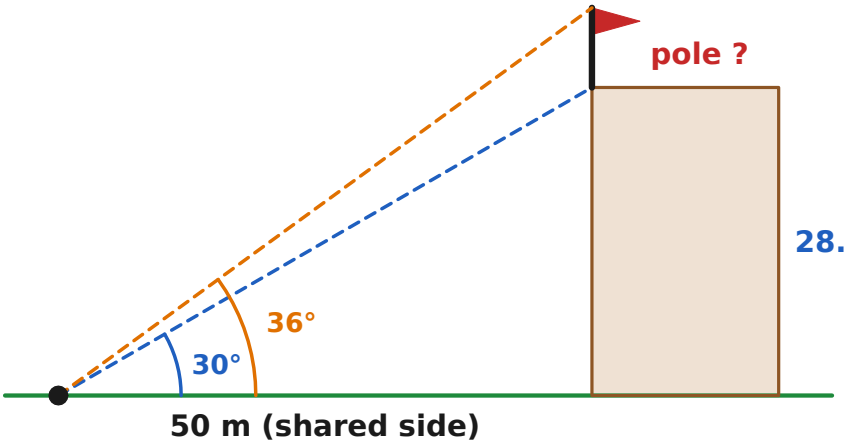
Example 2 (depression): from the top of a 60 m cliff, the angle of depression of a boat is 25°. How far is the boat from the foot of the cliff?

$$\tan 25^\circ = \frac{60}{d} \Rightarrow d = \frac{60}{\tan 25^\circ} = 128.67 \text{ m}$$

← d on the bottom: divide

(The 25° at the cliff top is also the angle of elevation from the boat, so it sits at the boat's corner of the triangle.)

Example 3 (two triangles): a flagpole stands on top of a building. From a point 50 m from the base, the angle of elevation of the top of the building is 30°, and of the top of the flagpole 36°. How tall is the flagpole?



Two right triangles share the 50 m side. Solve each with TOA, then subtract.

$$\text{building} = 50\tan 30^\circ = 28.87 \text{ m}$$ ← *smaller triangle*

$$\text{building} + \text{pole} = 50\tan 36^\circ = 36.33 \text{ m}$$ ← *bigger triangle, same 50 m*

$$\text{pole} = 36.33 - 28.87 = 7.46 \text{ m}$$ ← *subtract*

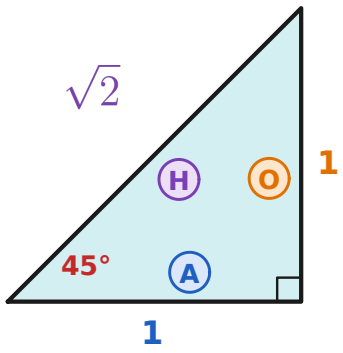
◿ Two special triangles give exact values

RULE: exact values

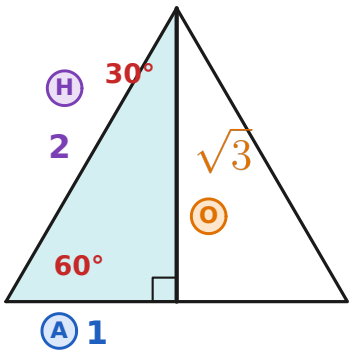
The $45°$ triangle has sides $1, 1, \sqrt{2}$. The $30°$–$60°$ triangle has sides $1, \sqrt{3}, 2$. Read sin, cos and tan off them with SOHCAHTOA (table below).

Useful patterns: $\sin 30° = \cos 60° = \frac{1}{2}$, $\sin 60° = \cos 30° = \frac{\sqrt{3}}{2}$, $\tan 45° = 1$.

A few angles have sin, cos and tan you can write exactly, with no calculator. They come from two triangles:



half a square



half an equilateral triangle (all sides 2)

badges from the 60° corner.
From the 30° corner, O and A swap:

opp = 1, adj = √3

Pythagoras gives the missing sides ($\sqrt{1^2 + 1^2} = \sqrt{2}$ and $\sqrt{2^2 - 1^2} = \sqrt{3}$). Now read off SOHCAHTOA:

	30°	45°	60°
sin	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
tan	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

Lecture 6 uses the 45° one: $(\cos 45°, \sin 45°) = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Side names	H opposite the right angle (longest); O across from θ; A next to θ
Pythagoras	$a^2 + b^2 = c^2$ c = hypotenuse. Hypotenuse: add then $\sqrt{}$. Short side: subtract then $\sqrt{}$
Pythagoras in 3D	$d^2 = a^2 + b^2 + c^2$ diagonal of a box
Circle	$(x-a)^2 + (y-b)^2 = r^2$ centre (a, b), radius r
SOHCAHTOA	$\sin\theta = \frac{O}{H}, \quad \cos\theta = \frac{A}{H}, \quad \tan\theta = \frac{O}{A}$ right-angled triangles only
Which ratio?	the one containing the side you know and the side you want
Rearranging	unknown on top: multiply; unknown on the bottom: divide; unknown angle: inverse
Inverse trig	$\theta = \sin^{-1}\frac{O}{H}, \ \cos^{-1}\frac{A}{H}, \ \tan^{-1}\frac{O}{A}$ one answer only: $\sin^{-1}, \tan^{-1}$ in $-90°$ to $90°$; $\cos^{-1}$ in $0°$ to $180°$
Elevation / depression	measured up / down from the **horizontal**; they are equal (parallel lines)
Two triangles	solve for the shared side first, then use it in the other triangle
Exact values	$\sin 30° = \frac{1}{2}, \ \sin 45° = \frac{1}{\sqrt{2}}, \ \sin 60° = \frac{\sqrt{3}}{2}$ cos: the same list backwards; $\tan 45° = 1$

◿ Your turn

Draw and label the triangle first (Lecture 6's advice). Answers in grey.

Pythagoras

1. Short sides 5 and 12: hypotenuse?
2. Hypotenuse 10, one side 6: the other side?
3. Diagonal of a 2 × 3 × 6 box

1. 13 2. 8 3. $\sqrt{(4 + 9 + 36)} = 7$

Sides

4. hyp 8, $\theta = 40°$: opposite?
5. adj 6, $\theta = 25°$: opposite?
6. opp 4, $\theta = 55°$: hypotenuse?

4. $8 \sin 40° \approx 5.14$ 5. $6 \tan 25° \approx 2.80$ 6. $4 \div \sin 55° \approx 4.88$

Angles

7. opp 5, hyp 13: θ?
8. adj 7, hyp 10: θ?
9. A ramp rises 1 m over 12 m of ground. Its angle?

7. $\sin^{-1}(5/13) \approx 22.6°$ 8. $\cos^{-1}(0.7) \approx 45.6°$ 9. $\tan^{-1}(1/12) \approx 4.8°$

Find the unknown

10. A 6 m ladder reaches 5 m up a wall. What angle does it make with the ground?
11. From 30 m away, the angle of elevation of a tree top is 28°. How tall is the tree?
12. From the top of a 45 m cliff, the angle of depression of a boat is 18°. How far is the boat from the foot of the cliff?
13. A ramp must rise 0.8 m at an angle of 5°. How long is the sloping surface?
14. $\tan \theta = 0.6$ and the adjacent side is 15. Find the opposite side and the hypotenuse.
15. From 40 m away, the elevation of the top of a building is 25° and of the top of a mast on its roof 31°. How tall is the mast?

10. $\sin \theta = 5/6$, $\theta \approx 56.4°$ 11. $30 \tan 28° \approx 15.95$ m 12. $45 \div \tan 18° \approx 138.50$ m 13. $\sin 5° = 0.8/H$, $H = 0.8 \div \sin 5° \approx 9.18$ m 14. $O = 0.6 \times 15 = 9$; $H = \sqrt{(81 + 225)} \approx 17.49$ 15. $40 \tan 31° - 40 \tan 25° \approx 24.03 - 18.65 = 5.38$ m